

Financial Econometrics

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Factor Models

CAPM

CAPM is a **one-factor model**, in which the net returns of all the assets are explained by the net return of the market portfolio.

The empirical failure of this model is not surprising :

- it relies on pretty unrealistic assumptions : for example, that investors have identical preferences, have the same information, and hold the same efficient portfolio.
- identifying and measuring the market return is difficult, if not impossible (the Roll Critique)

Multi-factor model

The empirical failure of the CAPM lead naturally to the introduction of **several explanatory factors**.

For instance, the assets of one country, or one industry, have specific risks in common.

The **Arbitrage Pricing Theory (APT)** was introduced by Ross (1976) in order to generalize the CAPM.

The APT specifies a pricing relationship with a number of "systematic" factors.

- 1 Model and Absence of Arbitrage Opportunities
- 2 Estimation with observable factors
- 3 Selection of Factors

APT

The APT model postulates a **linear relationship** between the **assets returns** and a finite number K of **factors**.

The theory is based on a **single period**.

$\mathbf{y} = (y_1, \dots, y_N)'$: vector of gross returns of N assets

$$y_i := y_{i,t+1} = \frac{p_{i,t+1}}{p_{i,t}}, \quad i = 1, \dots, N$$

$\mathbf{f} = (f_1, \dots, f_K)'$: vector of K common **underlying factors**. *

*. **Examples of factors** : the market portfolio (as in the CAPM) or some particular portfolios (for instance built with assets having a high balance sheet compared to their market value, see Fama and French (1992, 1995, 1996)), discrepancies between interest rates (e.g. between short-term and long-term rates), macroeconomic variables such as the inflation rate, the rate of growth of a country.

Factor model

A factor model writes

$$\begin{aligned}\mathbf{y} &= \mathbf{c} + \mathbf{B}\mathbf{f} + \mathbf{u} \\ E(\mathbf{u} | \mathbf{f}) &= \mathbf{0}, \quad E(\mathbf{u}\mathbf{u}' | \mathbf{f}) = \Sigma\end{aligned}$$

where $\mathbf{c} = (c_i)$ and $\mathbf{B} = (B_{i,j})$ are matrices of sizes $N \times 1$ and $N \times K$, Σ is a semi-positive definite.

The equation of asset i is thus

$$y_i = c_i + \mathbf{B}_i' \mathbf{f} + u_i$$

where $\mathbf{B}_i$ is the $K \times 1$ vector of **sensibilities (or betas)** of asset i to the factors.

Restrictions

Similar to the CAPM which entails $\alpha = 0$ in the regression model, the APT imposes restrictions to the previous model.

The APT relies on the assumption of *absence of arbitrage opportunity* (AAO) : a riskless portfolio yielding an expected gain superior to the riskless asset does not exist.

In other words, zero cost portfolio cannot have a strictly positive return.

Consequences

Suppose that $\Sigma = 0$, that is $\mathbf{u} = 0$ with probability 1.

Suppose that the law of the factors is non degenerate, that is

$\text{Var}(\mathbf{f})$ is **positive definite**.

A portfolio with composition $\mathbf{a} = (a_1, \dots, a_N)'$ in risky assets has return

$$\frac{\sum_{i=1}^N a_i p_{i,t+1}}{\sum_{j=1}^N a_j p_{j,t}} = \sum_{i=1}^N \frac{a_i p_{i,t}}{\sum_{j=1}^N a_j p_{j,t}} y_{i,t+1} = \bar{\mathbf{w}}_{\mathbf{a}}' \mathbf{y},$$

where $\bar{\mathbf{w}}_{\mathbf{a}}' = \left(\frac{a_1 p_{1t}}{\sum_{j=1}^N a_j p_{jt}}, \dots, \frac{a_N p_{Nt}}{\sum_{j=1}^N a_j p_{jt}} \right)$. Thus

$$\bar{\mathbf{w}}_{\mathbf{a}}' \mathbf{y} = \bar{\mathbf{w}}_{\mathbf{a}}' \mathbf{c} + \bar{\mathbf{w}}_{\mathbf{a}}' \mathbf{B} \mathbf{f}$$

and

$$E(\bar{\mathbf{w}}_{\mathbf{a}}' \mathbf{y}) = \bar{\mathbf{w}}_{\mathbf{a}}' \mathbf{c} + \bar{\mathbf{w}}_{\mathbf{a}}' \mathbf{B} E(\mathbf{f}), \quad \text{Var}(\bar{\mathbf{w}}_{\mathbf{a}}' \mathbf{y}) = \bar{\mathbf{w}}_{\mathbf{a}}' \mathbf{B} \text{Var}(\mathbf{f}) \mathbf{B}' \bar{\mathbf{w}}_{\mathbf{a}}.$$

Consequences (cont.)

Such a portfolio is riskless if $\text{Var}(\bar{\omega}'_a \mathbf{y}) = 0$, that is if $\bar{\omega}'_a \mathbf{B} = 0$ using the assumption of nondegenerate factors.

Thus, a riskless portfolio has expected return $\bar{\omega}'_a \mathbf{c}$.

The AAO assumption entails that

$$\forall \bar{\omega}_a \in \mathbb{R}^N, \quad \bar{\omega}'_a \mathbf{B} = 0 \quad \Rightarrow \quad \bar{\omega}'_a \mathbf{c} = \lambda_0$$

where λ_0 denotes the riskless asset return. Equivalently,

$$\forall \bar{\omega}_a \in \mathbb{R}^N \text{ such that } \bar{\omega}'_a \mathbf{e} = 1, \quad \bar{\omega}'_a \mathbf{B} = 0 \quad \Rightarrow \quad \bar{\omega}'_a (\mathbf{c} - \lambda_0 \mathbf{e}) = 0.$$

This implication means that $\mathbf{c} - \lambda_0 \mathbf{e}$ belongs to the space spanned by the columns of $\mathbf{B}$:

$$\mathbf{c} = \lambda_0 \mathbf{e} + \mathbf{B} \boldsymbol{\lambda}_1$$

where $\boldsymbol{\lambda}_1 \in \mathbb{R}^K$ is a vector of coefficients.

Consequences (cont.)

For asset i , we get

$$c_i = \lambda_0 + \sum_{j=1}^K B_{ij} \lambda_{1j}$$

where $\lambda_1 = (\lambda_{1j})$.

Finally, we find that the vector of net rentabilities $\mathbf{z} = \mathbf{y} - \lambda_0 \mathbf{e}$ must satisfy

$$\mathbf{z} = \mathbf{B}(\mathbf{f} + \lambda_1) + \mathbf{u}.$$

This formula is analogous to the equality $\alpha = 0$ in the CAPM.

General case - ICAPM

In the general case (when $\Sigma \neq 0$) the AAO no longer implies the previous equality for a fixed number of assets N , but the formula remains **valid asymptotically**, when $N \rightarrow \infty$ [†].

For fixed but large N the equality

$$\mathbf{c} = \lambda_0 \mathbf{e} + \mathbf{B}\lambda_1$$

becomes an approximation.

In an **intertemporal framework**, Merton (1973) obtained an exact formula (ICAPM, Intertemporal Capital Asset Pricing Model)

[†]. Assuming that the factors take into account all the systematic part of the risk of well-diversified portfolios (weight $1/N$ for each asset), the idiosyncratic part tends to 0. A crucial assumption for deriving the model is thus that the correlations of the components $\mathbf{u}$ be small in modulus.

Risk premia

$$\mathbf{c} = \lambda_0 \mathbf{e} + \mathbf{B}\boldsymbol{\lambda}_1 \Rightarrow E(\mathbf{y}) = \lambda_0 \mathbf{e} + \mathbf{B}(\boldsymbol{\lambda}_1 + E(\mathbf{f})).$$

Thus

$$\tilde{\boldsymbol{\mu}} = \mathbf{B}\boldsymbol{\lambda}^*$$

where $\tilde{\boldsymbol{\mu}} = \boldsymbol{\mu} - \lambda_0 \mathbf{e}$ and $\boldsymbol{\lambda}^* = \boldsymbol{\lambda}_1 + E(\mathbf{f})$.

The components λ_i^* of $\boldsymbol{\lambda}^*$ can be interpreted as the *risk premia* associated with the K factors.

The risk premium of each asset is a linear combination of the risk premia associated with the factors.

Can this formula be [tested empirically](#)? the theory does not tell what the factors are.

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System of stacked regressions

In this section, we assume the factors are **known**.

Example : the three-factor model of Fama-French (1992)

- the excess return of a market portfolio
- the difference between the return on a portfolio of small stocks and the return on a portfolio of large stocks (SMB, small minus big)
- the difference between the return on a portfolio of high-book-to-market stocks and the return on a portfolio of low-book-to-market stocks (HML, high minus low)

The i -th equation has the form

$$y_{it} = c_i + B_{i1}y_t^{(m)} + B_{i2}\text{SMB}_t + B_{i3}\text{HML}_t + u_{it}, \quad t = 1, \dots, n$$

System of stacked regressions

The general model writes

$$\mathbf{y}_t = \mathbf{c} + \mathbf{B}\mathbf{f}_t + \mathbf{u}_t, \quad E(\mathbf{u}_t | \mathbf{f}_t) = \mathbf{0}, \quad E(\mathbf{u}_t \mathbf{u}_t' | \mathbf{f}_t) = \mathbf{\Sigma}$$

where $\mathbf{y}_t \in \mathbb{R}^N$, $\mathbf{f}_t \in \mathbb{R}^K$, $\mathbf{B} = (B_{i,j})$ is a $N \times K$ matrix of factor loadings, $\mathbf{c} \in \mathbb{R}^N$.

The unknown parameters are the coefficients of the matrices $\mathbf{B}$ and $\mathbf{c}$, and the variance-covariance matrix $\mathbf{\Sigma}$.

The observations are $(\mathbf{y}_t, \mathbf{f}_t)$, $t = 1, \dots, n$.

Estimation method

As in the CAPM, we have a system of stacked regressions with the same explicative variables $1, f_{1t}, \dots, f_{Kt}$ in each equation.

The parameters can thus be **estimated equation by equation**, by LS.

The equation for asset i writes

$$y_{it} = c_i + B_{i,1}f_{1t} + \dots + B_{i,K}f_{Kt} + u_{it}.$$

For convenience we consider ML estimators which, in the Gaussian case, provide the **same estimators** as the LS.

ML estimators

Proposition

If the law of $\mathbf{u}_t$ conditional on the $\mathbf{f}_t$ is Gaussian, $\mathcal{N}(\mathbf{0}, \Sigma)$, the ML estimator of $\mathbf{c}, \mathbf{B}$ and Σ is given by

$$\hat{\mathbf{c}}_n = \bar{\mathbf{y}}_n - \hat{\mathbf{B}}_n \bar{\mathbf{f}}_n,$$

$$\hat{\mathbf{B}}_n = \left(\frac{1}{n} \sum_{t=1}^n (\mathbf{y}_t - \bar{\mathbf{y}}_n)(\mathbf{f}_t - \bar{\mathbf{f}}_n)' \right) \left(\frac{1}{n} \sum_{t=1}^n (\mathbf{f}_t - \bar{\mathbf{f}}_n)(\mathbf{f}_t - \bar{\mathbf{f}}_n)' \right)^{-1}$$

$$\hat{\Sigma}_n = \frac{1}{n} \sum_{t=1}^n (\mathbf{y}_t - \hat{\mathbf{c}}_n - \hat{\mathbf{B}}_n \mathbf{f}_t)(\mathbf{y}_t - \hat{\mathbf{c}}_n - \hat{\mathbf{B}}_n \mathbf{f}_t)'$$

provided that the square matrix involved in $\hat{\mathbf{B}}_n$ be non-singular.

Proof

The log-likelihood writes

$$\begin{aligned} L(\mathbf{c}, \mathbf{B}, \mathbf{\Sigma}) &= -\frac{Nn}{2} \log(2\pi) - \frac{n}{2} \log|\mathbf{\Sigma}| \\ &\quad - \frac{1}{2} \sum_{t=1}^n (\mathbf{y}_t - \mathbf{c} - \mathbf{B}\mathbf{f}_t)' \mathbf{\Sigma}^{-1} (\mathbf{y}_t - \mathbf{c} - \mathbf{B}\mathbf{f}_t). \end{aligned}$$

Maximizing $L(\cdot)$ thus amounts to minimizing

$$L^*(\mathbf{c}, \mathbf{B}, \mathbf{\Sigma}) = n \log|\mathbf{\Sigma}| + \text{Tr} \left\{ \mathbf{\Sigma}^{-1} \sum_{t=1}^n (\mathbf{y}_t - \mathbf{c} - \mathbf{B}\mathbf{f}_t)(\mathbf{y}_t - \mathbf{c} - \mathbf{B}\mathbf{f}_t)' \right\}$$

Proof

The parameters $\mathbf{c}$ and $\mathbf{B}$ only appear in the last term, which can be written, after centering :

(using $\frac{1}{n} \sum_{t=1}^n X_t X_t' = \frac{1}{n} \sum_{t=1}^n (X_t - \bar{X}_n)(X_t - \bar{X}_n)' + \bar{X}_n \bar{X}_n'$)

$$\begin{aligned}
 & \text{Tr} \left\{ \Sigma^{-1} \sum_{t=1}^n \{ \mathbf{y}_t - \bar{\mathbf{y}}_n - \mathbf{B}(\mathbf{f}_t - \bar{\mathbf{f}}_n) \} \{ \mathbf{y}_t - \bar{\mathbf{y}}_n - \mathbf{B}(\mathbf{f}_t - \bar{\mathbf{f}}_n) \}' \right\} \\
 & + n \text{Tr} \left\{ \Sigma^{-1} (\bar{\mathbf{y}}_n - \mathbf{c} - \mathbf{B} \bar{\mathbf{f}}_n) (\bar{\mathbf{y}}_n - \mathbf{c} - \mathbf{B} \bar{\mathbf{f}}_n)' \right\} \\
 = & \text{Tr} \left\{ \Sigma^{-1} \sum_{t=1}^n \{ \mathbf{y}_t - \bar{\mathbf{y}}_n - \hat{\mathbf{B}}_n(\mathbf{f}_t - \bar{\mathbf{f}}_n) \} \{ \mathbf{y}_t - \bar{\mathbf{y}}_n - \hat{\mathbf{B}}_n(\mathbf{f}_t - \bar{\mathbf{f}}_n) \}' \right\} \\
 & + \text{Tr} \left\{ \Sigma^{-1} (\hat{\mathbf{B}}_n - \mathbf{B}) \sum_{t=1}^n (\mathbf{f}_t - \bar{\mathbf{f}}_n) (\mathbf{f}_t - \bar{\mathbf{f}}_n)' (\hat{\mathbf{B}}_n - \mathbf{B})' \right\} \\
 & + n \text{Tr} \left\{ \Sigma^{-1} (\bar{\mathbf{y}}_n - \mathbf{c} - \mathbf{B} \bar{\mathbf{f}}_n) (\bar{\mathbf{y}}_n - \mathbf{c} - \mathbf{B} \bar{\mathbf{f}}_n)' \right\}.
 \end{aligned}$$

Proof

This expression is positive whatever $\mathbf{B}$ and $\mathbf{c}$, and the last 2 terms cancel only when $\mathbf{B} = \hat{\mathbf{B}}_n$ and $\mathbf{c} = \hat{\mathbf{c}}_n$. Thus $(\hat{\mathbf{c}}_n, \hat{\mathbf{B}}_n)$ is the ML estimator of $(\mathbf{c}, \mathbf{B})$.

It follows that

$$\begin{aligned} L^*(\hat{\mathbf{c}}_n, \hat{\mathbf{B}}_n, \Sigma) &= n \log |\Sigma| + n \text{Tr}(\Sigma^{-1} \hat{\Sigma}_n) \\ &= n \log |\hat{\Sigma}_n| + n \{ \text{Tr}(\Sigma^{-1} \hat{\Sigma}_n) - \log |\Sigma^{-1} \hat{\Sigma}_n| \} \\ &\geq n \log |\hat{\Sigma}_n| + nN = L^*(\hat{\mathbf{c}}_n, \hat{\mathbf{B}}_n, \hat{\Sigma}_n). \end{aligned}$$

[For any symmetric matrices A and B , where A is pos.def. and B is semi-pos.def., $\text{Tr}(A^{-1}B) - \log|A^{-1}B|$ is minimal for $A=B$.] ► Proof

The likelihood is thus maximal at $(\hat{\mathbf{c}}_n, \hat{\mathbf{B}}_n, \hat{\Sigma}_n)$.

□

Remark

If the $\mathbf{f}_t$ are independent and such that $\text{Var}(\mathbf{f}_t) := \Gamma$ is a positive-definite matrix, the invertibility condition is satisfied.

Indeed, otherwise we would have, for some vector $\mathbf{x} \neq 0$,

$$\begin{aligned} 0 &= \mathbf{x}' \left(\frac{1}{n} \sum_{t=1}^T (\mathbf{f}_t - \bar{\mathbf{f}}_n)(\mathbf{f}_t - \bar{\mathbf{f}}_n)' \right) \mathbf{x} \\ &= \frac{1}{n} \sum_{t=1}^T \left(\mathbf{x}'(\mathbf{f}_t - \bar{\mathbf{f}}_n) \right)^2 \end{aligned}$$

thus

$$\mathbf{x}'\mathbf{f}_t = \mathbf{x}'\bar{\mathbf{f}}_n.$$

By taking the variance of each side of this equality, we get

$$\mathbf{x}'\Gamma\mathbf{x} = \mathbf{x}'\text{Var}(\bar{\mathbf{f}}_n)\mathbf{x} = \left(\frac{1}{n}\right)\mathbf{x}'\Gamma\mathbf{x}.$$

Hence $\mathbf{x}'\Gamma\mathbf{x} = 0$, which is not possible since Γ is non-singular.

Asymptotic distribution of the ML estimator

$$\mathbf{y}_t = \mathbf{c} + \mathbf{B}\mathbf{f}_t + \mathbf{u}_t, \quad E(\mathbf{u}_t | \mathbf{f}_t) = \mathbf{0}, \quad E(\mathbf{u}_t \mathbf{u}_t' | \mathbf{f}_t) = \Sigma$$

Assume that

- $\bar{\mathbf{f}}_n \rightarrow E(\mathbf{f}_t) := \mathbf{\Lambda}$ and $\frac{1}{n} \sum_{t=1}^n (\mathbf{f}_t - \bar{\mathbf{f}}_n)(\mathbf{f}_t - \bar{\mathbf{f}}_n)' \rightarrow \mathbf{\Gamma}$ in probability, where $\mathbf{\Gamma}$ is positive definite.
- A CLT applies for the sequence $(\mathbf{f}_t, \mathbf{u}_t)$ (details in the proof)

Proposition

$$\sqrt{n} \begin{pmatrix} \hat{\mathbf{c}}_n - \mathbf{c} \\ \text{vec}(\hat{\mathbf{B}}_n - \mathbf{B}) \end{pmatrix} \xrightarrow{d} \mathcal{N} \left(0, \begin{pmatrix} (1 + \mathbf{\Lambda}' \mathbf{\Gamma}^{-1} \mathbf{\Lambda}) \Sigma & -\mathbf{\Lambda}' \mathbf{\Gamma}^{-1} \otimes \Sigma \\ -\mathbf{\Gamma}^{-1} \mathbf{\Lambda} \otimes \Sigma & \mathbf{\Gamma}^{-1} \otimes \Sigma \end{pmatrix} \right).$$

Sketch of the proof :

$$\begin{aligned}
 \mathbf{y}_t &= \mathbf{c} + \mathbf{B}\mathbf{f}_t + \mathbf{u}_t & \Rightarrow & \quad \bar{\mathbf{y}}_n = \mathbf{c} + \mathbf{B}\bar{\mathbf{f}}_n + \bar{\mathbf{u}}_n \\
 & & \Rightarrow & \quad \mathbf{y}_t - \bar{\mathbf{y}}_n = \mathbf{B}(\mathbf{f}_t - \bar{\mathbf{f}}_n) + \mathbf{u}_t - \bar{\mathbf{u}}_n
 \end{aligned}$$

$$\begin{aligned}
 \sqrt{n}(\hat{\mathbf{B}}_n - \mathbf{B}) &= \left(\frac{1}{\sqrt{n}} \sum_{t=1}^n (\mathbf{u}_t - \bar{\mathbf{u}}_n)(\mathbf{f}_t - \bar{\mathbf{f}}_n)' \right) \left(\frac{1}{n} \sum_{t=1}^n (\mathbf{f}_t - \bar{\mathbf{f}}_n)(\mathbf{f}_t - \bar{\mathbf{f}}_n)' \right)^{-1} \\
 &\stackrel{d}{=} \left(\frac{1}{\sqrt{n}} \sum_{t=1}^n (\mathbf{u}_t - \bar{\mathbf{u}}_n) \underbrace{(\mathbf{f}_t - \bar{\mathbf{f}}_n)'}_{\tilde{\mathbf{f}}_t} \right) \Gamma^{-1} = \left(\frac{1}{\sqrt{n}} \sum_{t=1}^n \mathbf{u}_t \tilde{\mathbf{f}}_t' \right) \Gamma^{-1},
 \end{aligned}$$

$$\begin{aligned}
 \Rightarrow \quad \sqrt{n} \operatorname{vec}(\hat{\mathbf{B}}_n - \mathbf{B}) &= (\Gamma^{-1} \otimes \mathbf{I}_N) \left(\frac{1}{\sqrt{n}} \sum_{t=1}^n \operatorname{vec}(\mathbf{u}_t \tilde{\mathbf{f}}_t') \right) \\
 &= (\Gamma^{-1} \otimes \mathbf{I}_N) \left(\frac{1}{\sqrt{n}} \sum_{t=1}^n \tilde{\mathbf{f}}_t \otimes \mathbf{u}_t \right).
 \end{aligned}$$

Sketch of the proof (cont.) :

$$\begin{aligned}
\mathbf{y}_t &= \mathbf{c} + \mathbf{B}\mathbf{f}_t + \mathbf{u}_t \\
\Rightarrow \quad \bar{\mathbf{y}}_n &= \mathbf{c} + \mathbf{B}\bar{\mathbf{f}}_n + \bar{\mathbf{u}}_n = \hat{\mathbf{c}}_n + \hat{\mathbf{B}}_n \bar{\mathbf{f}}_n \\
\Rightarrow \quad \hat{\mathbf{c}}_n - \mathbf{c} &= \bar{\mathbf{u}}_n - (\hat{\mathbf{B}}_n - \mathbf{B})\bar{\mathbf{f}}_n = \bar{\mathbf{u}}_n - \underbrace{(\bar{\mathbf{f}}_n' \otimes \mathbf{I}_N) \text{vec}(\hat{\mathbf{B}}_n - \mathbf{B})}_{(\Gamma^{-1} \otimes \mathbf{I}_N) \left(\frac{1}{n} \sum_{t=1}^n \tilde{\mathbf{f}}_t \otimes \mathbf{u}_t \right)}
\end{aligned}$$

Thus

$$\sqrt{n} \begin{pmatrix} \hat{\mathbf{c}}_n - \mathbf{c} \\ \text{vec}(\hat{\mathbf{B}}_n - \mathbf{B}) \end{pmatrix} = \begin{pmatrix} \mathbf{I}_N & -E(\mathbf{f}_t)' \Gamma^{-1} \otimes \mathbf{I}_N \\ 0 & \Gamma^{-1} \otimes \mathbf{I}_N \end{pmatrix} \begin{pmatrix} \frac{1}{\sqrt{n}} \sum_{t=1}^n \mathbf{u}_t \\ \frac{1}{\sqrt{n}} \sum_{t=1}^n \tilde{\mathbf{f}}_t \otimes \mathbf{u}_t \end{pmatrix} := \mathbf{A} \mathbf{X}_n.$$

Moreover, since $E(\tilde{\mathbf{f}}_t \otimes \mathbf{u}_t)(\tilde{\mathbf{f}}_t \otimes \mathbf{u}_t)' = E(\tilde{\mathbf{f}}_t \tilde{\mathbf{f}}_t' \otimes \mathbf{u}_t \mathbf{u}_t') = \Gamma \otimes \Sigma$, if a CLT applies,

$$\mathbf{X}_n \xrightarrow{d} \mathcal{N} \left(0, \begin{pmatrix} \Sigma & 0 \\ 0 & \Gamma \otimes \Sigma \end{pmatrix} \right).$$

□

Tests

- The **absence of an intercept** $H_0: \mathbf{c} = \mathbf{0}$, can be tested using the Wald statistic (if Σ is non-singular)

$$W = \left(1 + \hat{\Lambda}' \hat{\Gamma}^{-1} \hat{\Lambda}\right)^{-1} \hat{\mathbf{c}}_n' \hat{\Sigma}^{-1} \hat{\mathbf{c}}_n \xrightarrow{d} \chi_N^2 \quad (\text{under } H_0).$$

- The **AAO restriction characterizing the APT** can be tested, for instance using a **Likelihood Ratio test**. In the model constrained by

$$\mathbf{c} = \lambda_0 \mathbf{e} + \mathbf{B} \boldsymbol{\lambda}_1,$$

the parameters are the coefficients of the matrices $\mathbf{B}$ and Σ , the estimation of $\mathbf{c} \in \mathbb{R}^N$ being replaced by that of $\lambda_0 \in \mathbb{R}$ and $\boldsymbol{\lambda}_1 \in \mathbb{R}^K$.

The LR statistic asymptotically follows under H_0 a $\chi^2(N - K - 1)$.

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- Principal Components Analysis
- Factor Analysis
- Application

Unknown factors

We no longer assume that the factors are known.

Two main statistical approaches for determining the factors are :

- the Principal Components Analysis (PCA)
- the Factor Analysis.

PCA

PCA is a general technique aiming at **reducing the number of variables** without losing too much information on the covariance structure.

In our setting, the aim is to reduce the dimension N of the vector $\mathbf{y}_t$ of the individual returns.

We will look for a **limited number of linear combinations** of the baseline returns (the **principal components**) allowing to **approximate the covariance matrix** $\mathbf{\Omega}_t$, or the correlation matrix $\mathbf{\rho}_t$.

In this approach, the factors are the principal components.

First principal component

Let $\mathbf{a}$ and $\mathbf{b}$ two vectors of $\mathbb{R}^N$, which are non random conditional on the available information at time t .

We have

$$\text{Var}_t(\mathbf{a}'\mathbf{y}_{t+1}) = \mathbf{a}'\boldsymbol{\Omega}_t\mathbf{a}, \quad \text{Cov}_t(\mathbf{a}'\mathbf{y}_{t+1}, \mathbf{b}'\mathbf{y}_{t+1}) = \mathbf{a}'\boldsymbol{\Omega}_t\mathbf{b}.$$

The **first principal component** is the (normalized) linear combination $\mathbf{a}'_1\mathbf{y}_t$ with the largest variance.

We thus obtain $\mathbf{a}_1$ by solving

$$\begin{cases} \max_{\mathbf{a}} \mathbf{a}'\boldsymbol{\Omega}_t\mathbf{a} \\ \text{s.t.} \quad \mathbf{a}'\mathbf{a} = 1. \end{cases}$$

Other principal components

The **2nd principal component** is the (normalized) linear combination $\mathbf{a}'_2 \mathbf{y}_t$ with the largest variance and such that $\mathbf{a}_2$ is orthogonal to $\mathbf{a}_1$.

We thus obtain $\mathbf{a}_2$ by solving

$$\begin{cases} \max_{\mathbf{a}} \mathbf{a}' \boldsymbol{\Omega}_t \mathbf{a} \\ \text{subject to} & \mathbf{a}' \mathbf{a} = 1, \quad \mathbf{a}' \mathbf{a}_1 = 0. \end{cases}$$

More generally, we obtain the **i th principal component** $\mathbf{a}'_i \mathbf{y}_t$ (for $(i > 1)$) by solving

$$\begin{cases} \max_{\mathbf{a}} \mathbf{a}' \boldsymbol{\Omega}_t \mathbf{a} \\ \text{subject to} & \mathbf{a}' \mathbf{a} = 1, \quad \mathbf{a}' \mathbf{a}_j = 0, \quad j = 1, \dots, i-1. \end{cases}$$

Spectral decomposition of the variance-covariance matrix

Since $\mathbf{\Omega}_t$ is a positive-definite matrix, there exists a $N \times N$ orthogonal matrix $\mathbf{P}$ (i.e. $\mathbf{P}'\mathbf{P} = \mathbf{I}_N$), built with the eigenvectors of $\mathbf{\Omega}_t$, such that $\mathbf{P}'\mathbf{\Omega}_t\mathbf{P} = \mathbf{\Lambda}$, where $\mathbf{\Lambda} = \text{diag}(\lambda_1, \dots, \lambda_N)$ and

$$\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_N > 0.$$

Let $\mathbf{e}_i$ the column-vector of $\mathbf{P}$ associated with the eigenvalue λ_i :

$$\mathbf{P} = [\mathbf{e}_1 \dots \mathbf{e}_N], \quad \text{with} \quad \mathbf{e}_i' \mathbf{e}_i = 1, \quad \mathbf{e}_i' \mathbf{e}_j = 0 \quad (j \neq i).$$

The spectral decomposition of $\mathbf{\Omega}_t = \mathbf{P}\mathbf{\Lambda}\mathbf{P}'$ is

$$\mathbf{\Omega}_t = \sum_{i=1}^N \lambda_i \mathbf{e}_i \mathbf{e}_i'.$$

Thus

$$\mathbf{a}' \mathbf{\Omega}_t \mathbf{a} = \sum_{i=1}^N \lambda_i (\mathbf{a}' \mathbf{e}_i)^2.$$

Variance decomposition

Thus, the i th principal component of $\mathbf{y}_{t+1}$ is

$$r_{i,t+1} = \mathbf{e}_i' \mathbf{y}_{t+1},$$

pour $i = 1, \dots, N$.

When the eigenvalues are distinct, the eigenvectors are uniquely defined (up to the sign). If some eigenvalues are equal, the principal components are not uniquely defined.

We have

$$\text{Var}_t(r_{i,t+1}) = \lambda_i, \quad \text{Cov}_t(r_{i,t+1}, r_{j,t+1}) = 0 \quad (i \neq j),$$

and thus

$$\sum_{i=1}^N \text{Var}_t(y_{i,t+1}) = \text{tr}(\mathbf{\Omega}_t) = \sum_{i=1}^N \lambda_i = \sum_{i=1}^N \text{Var}_t(r_{i,t+1}).$$

Variance decomposition

Thus the **part of the total variance** which is explained by the i th principal component is

$$\frac{\text{Var}_t(r_{i,t+1})}{\sum_{i=1}^N \text{Var}_t(y_{i,t+1})} = \frac{\lambda_i}{\lambda_1 + \dots + \lambda_N}.$$

In practice, the aim is to select a small rank i such that the **cumulated proportion of the total variance**

$$\frac{\lambda_1 + \dots + \lambda_i}{\lambda_1 + \dots + \lambda_N} \quad \text{be large.}$$

If we work with the correlation matrix rather than the covariance matrix, we have $\text{tr}(\boldsymbol{\rho}_t) = N$ and the cumulated proportion is

$$(\lambda_1 + \dots + \lambda_i)/N.$$

In practice

$\mathbf{\Omega}_t$ is estimated by $\hat{\mathbf{\Omega}}_n$ and estimators of the eigenvalues and principal components follow.

The **spectral decomposition** of $\hat{\mathbf{\Omega}}_n$ is

$$\hat{\mathbf{\Omega}}_n = \sum_{i=1}^N \hat{\lambda}_i \hat{\mathbf{e}}_i \hat{\mathbf{e}}_i'$$

where the $\hat{\mathbf{e}}_i' \mathbf{y}_{t+1}$, are called **empirical principal components** of $\mathbf{y}_{t+1}$.

How many significant principal components should be kept?

Several **informal and graphical methods** can be used (for instance the "**scree plot**" consisting in drawing the eigenvalues by decreasing order of modulus in order to detect a break).

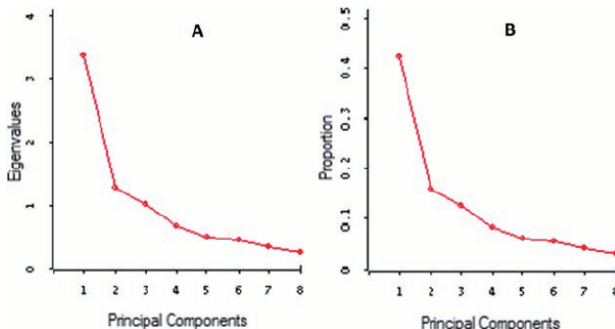


Figure – Scree plot representing the eigenvalues (A) and the part of total variance which is explained by the i th principal component (B).

Behaviour of estimated eigenvalues in large dimension

Asymptotic results on the distributions of the eigenvalues and eigenvectors of matrices exist, when both N and n go to ∞ .

Asymptotic setting : Both n and $N = N(n)$ tend to ∞ such that

$$\frac{N(n)}{n} \longrightarrow c \in (0, \infty).$$

For an empirical variance of the form $\mathbf{S} = \frac{1}{n} \sum_{t=1}^n \mathbf{X}_t \mathbf{X}_t'$, $\mathbf{X}_t \in \mathbb{R}^N$, results are known when

$$(X_{i,t})_{i,t} \text{ is i.i.d., } E(X_{i,t}) = 0, \quad E(X_{i,t}^2) = 1, \quad E(X_{i,t}^4) < \infty.$$

We thus have $E(\mathbf{X}_t \mathbf{X}_t') = I_N$: the theoretical eigenvalues are equal to 1.

Empirical eigenvalues

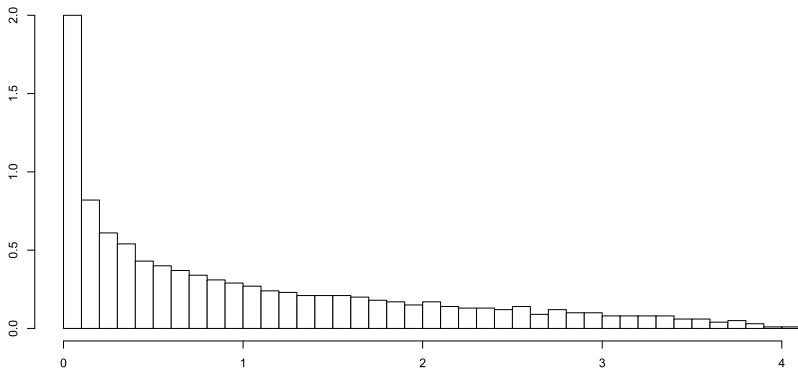
The empirical eigenvalues are known to be **much more dispersed** than the theoretical values.

Example :

- $n = 10$ draws of a random vector $\mathbf{X} \sim \mathcal{N}(\mathbf{0}, \mathbf{\Sigma} = \mathbf{I}_{10})$
- The theoretical eigenvalues are all equal to 1.
- There is an **extreme spread** of the eig. $\hat{\lambda}_i$ of $\frac{1}{10} \sum_{t=1}^{10} \mathbf{X}_t \mathbf{X}_t'$:
a typical example is
3.07, 1.40, 1.12, 0.78, 0.51, 0.30, 0.16, 0.095, 0.036, 0.003.
- One could conclude from the sample that the true eigenvalues are different from each other.

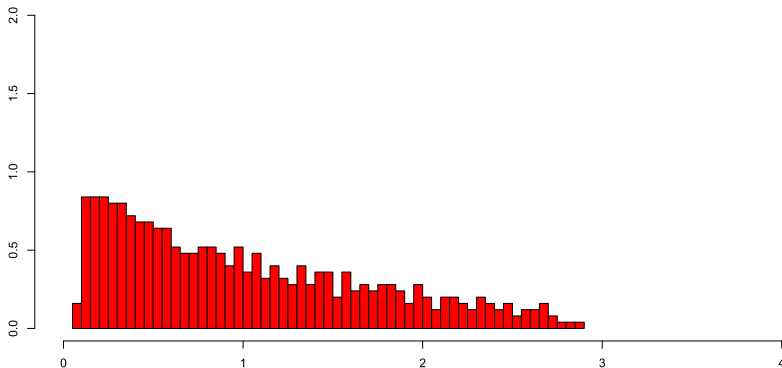
$n = 1000, N = 1000$

THE SAMPLE COVARIANCE MATRIX SPECTRUM



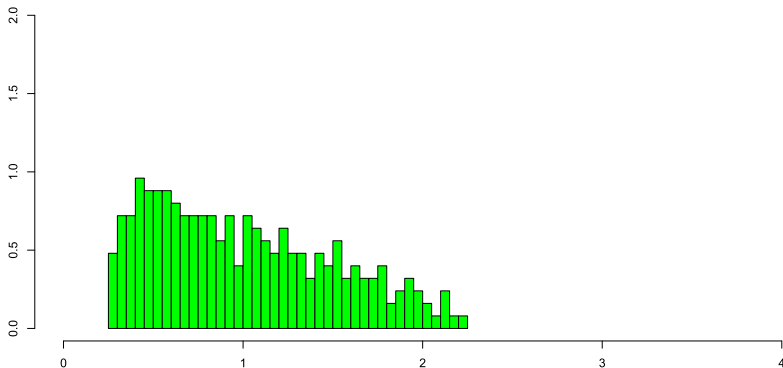
$n = 1000, N = 500$

THE SAMPLE COVARIANCE MATRIX SPECTRUM



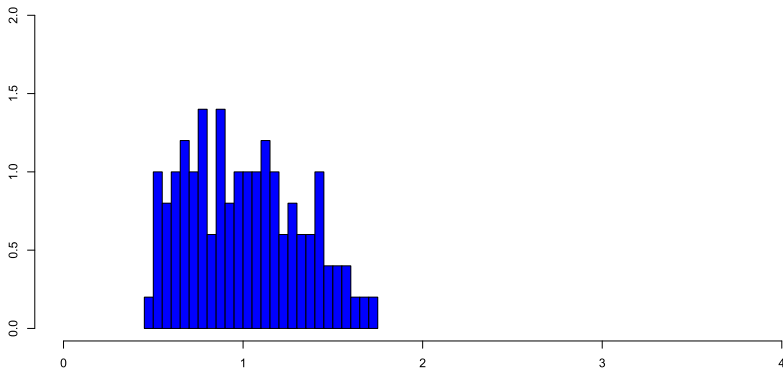
$n = 1000, N = 250$

THE SAMPLE COVARIANCE MATRIX SPECTRUM



$n = 1000, N = 100$

THE SAMPLE COVARIANCE MATRIX SPECTRUM



Empirical spectral distribution of $\mathbf{S}$ in large dimension

$$\hat{F}(\lambda) = \frac{1}{N} \sum_{i=1}^N \mathbf{1}_{\hat{\lambda}_i \leq \lambda}, \quad \hat{\lambda}_1 \geq \dots \geq \hat{\lambda}_N \text{ eigenvalues of } \mathbf{S}$$

$\hat{F}$ converges a.s. to a non-random distribution F_c .

- If $c \leq 1$, this distribution is continuous, with density (Marčenko-Pastur (1967) law)

$$f_c(\lambda) = \frac{1}{2\pi c} \sqrt{\frac{(b-\lambda)(\lambda-a)}{\lambda^2}} \mathbf{1}_{a \leq \lambda \leq b}$$

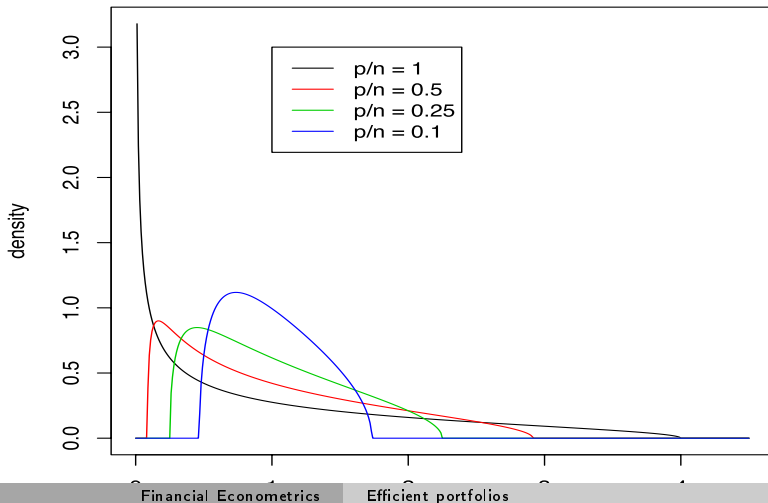
where $a = (1 - \sqrt{c})^2$ and $b = (1 + \sqrt{c})^2$.

When $c = 0$, the largest and smallest eigenvalues converge to 1.

- If $c > 1$, the asymptotic distribution is a mixture of a point mass at 0 (weight $1 - 1/c$) and the density $f_{1/c}$ (weight $1/c$)

Asymptotic distribution of the empirical eigenvalues

Density functions of Marcenko–Pastur law



Extension : Spiked covariance model (Johnstone, 2001)

The previous results concerned covariance matrices with identical eigenvalues.

Suppose now that the spectral decomposition of $\mathbf{\Omega}$ is

$$\mathbf{\Omega} = \sum_{j=1}^M \lambda_j \mathbf{e}_j \mathbf{e}_j' + \mathbf{I}_N, \quad \lambda_1 \geq \dots \geq \lambda_M.$$

We thus have M eigenvalues $\ell_i = \lambda_i + 1$, the remaining ones being equal to 1.

When $N/n \rightarrow c$ and for i.i.d. observations, for $j = 1, \dots, M$

$$\hat{\ell}_j \xrightarrow{a.s.} \begin{cases} (1 + \sqrt{c})^2 & \text{if } \ell_j \leq 1 + \sqrt{c} \\ \ell_j \left(1 + \frac{c}{\ell_j - 1}\right) & \text{if } \ell_j > 1 + \sqrt{c} \end{cases}$$

Spiked covariance model estimation

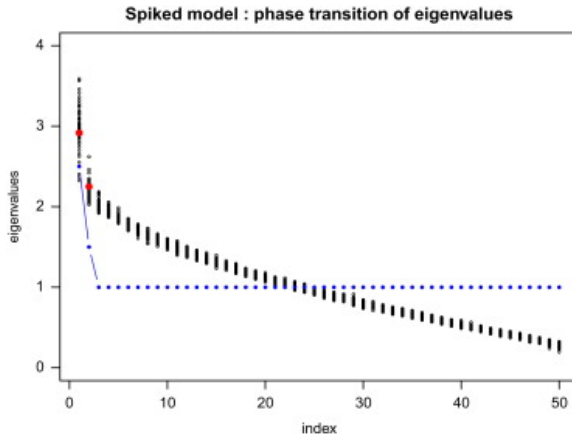


Figure – Eigenvalues in a spiked covariance model : $N=50$, $n=200$ and eigenvalues of the covariance matrix are $\ell_1=2.5$, $\ell_2=1.5$ and $\ell_j=1$ for $j=3,\dots,N$. So $\ell_1 > 1 + \sqrt{\frac{N}{n}}$ and $\ell_2 = 1 + \sqrt{\frac{N}{n}}$. Blue dots = population eigenvalues. Black circles = sample eigenvalues (based on i.i.d. Gaussian samples) for 50 replicates. Solid red circles = the theoretical limits of the first two eigenvalues for $\frac{N}{n}=0.25$

Factor Analysis

While PCA focuses on the **covariance structure** of the variables, the factor analysis aims at **explaining interdependencies** between a set of variables from a **small number of unobservable factors**.

The model has the form

$$\mathbf{y}_t = \mathbf{c} + \mathbf{B}\mathbf{f}_t + \mathbf{u}_t, \quad E(\mathbf{u}_t | \mathbf{f}_t) = \mathbf{0}, \quad E(\mathbf{u}_t \mathbf{u}_t' | \mathbf{f}_t) = \Sigma$$

but the factors $\mathbf{f}_t$ are now **unobservable**.

It is not restrictive to assume $E(\mathbf{f}_t) = \mathbf{0}$: otherwise $E(\mathbf{y}_t) = \mathbf{c} + \mathbf{B}E(\mathbf{f}_t)$ and the intercept $\mathbf{c}$ can be modified.

Let $\mathbf{\Gamma} = \text{Var}(\mathbf{f}_t)$. The variance-covariance matrix of $\mathbf{y}_t$ writes

$$\mathbf{\Omega} = \mathbf{B}\mathbf{\Gamma}\mathbf{B}' + \Sigma.$$

Identifiability of the factors

Without additional restriction, the factors are **unidentified** : if $\mathbf{C}$ is a $K \times K$ invertible matrix and if we let $\mathbf{f}_t^* = \mathbf{C}^{-1}\mathbf{f}_t$, $\mathbf{B}^* = \mathbf{B}\mathbf{C}$, the model can be written

$$\mathbf{y}_t = \mathbf{c} + \mathbf{B}^* \mathbf{f}_t^* + \mathbf{u}_t.$$

It is thus not possible to estimate the **factor loadings matrix** $\mathbf{B}$ and the factors $\mathbf{f}_t$ without further constraints.

A way to solve - partially - the indetermination is to impose

$$E(\mathbf{f}_t \mathbf{f}_t') = \mathbf{I}_K, \quad \text{i.e.} \quad \mathbf{f}_{it}^2 = 1, \quad \mathbf{f}_{it} \mathbf{f}_{jt} = 0 \quad (j \neq i).$$

where $\mathbf{I}_K$ is the $K \times K$ identity matrix. Indeed we thus have $E(\mathbf{f}_t^* \mathbf{f}_t^{*'}) = \mathbf{C}^{-1}(\mathbf{C}^{-1})' = \mathbf{I}_K$. The factors are called **orthogonal**.

Orthogonal factors

The orthogonal factors are only determined **up to a rotation** (multiplication by an orthogonal matrix).

This **non-uniqueness of the factors** is a weakness of the factor analysis because the factors are arbitrary.

It can also be an **advantage** because when it allows to choose factors which can be **interpreted**.

Orthogonal factors

Under the assumption of orthogonal factors, the variance-covariance matrix of $\mathbf{y}_t$ writes

$$\mathbf{\Omega} = \mathbf{B}\mathbf{B}' + \mathbf{\Sigma}.$$

In particular, we have

$$\text{Var}(y_{it}) = B_{i1}^2 + \dots B_{iK}^2 + \text{Var}(\epsilon_{it}).$$

Thus,

$$B_{i1}^2 + \dots B_{iK}^2$$

can be interpreted as the part of the variance of y_{it} which is explained by the factors.

Identifiability of the parameters

Problem : For a given matrix $\mathbf{\Omega}$ and a number K of factors, find a triplet $(\mathbf{B}, \mathbf{\Gamma}, \mathbf{\Sigma})$ with $\mathbf{\Sigma}$ **diagonal** such that

$$\mathbf{\Omega} = \mathbf{B}\mathbf{\Gamma}\mathbf{B}' + \mathbf{\Sigma}.$$

When such a triplet **exists and is unique**, the model is called **identifiable**.

If we assume that the factors are orthogonal, the problem is to find $(\mathbf{B}, \mathbf{\Sigma})$ such that

$$\mathbf{\Omega} = \mathbf{B}\mathbf{B}' + \mathbf{\Sigma}.$$

It can be easier to assume that $\mathbf{\Gamma}_0 := \mathbf{B}'\mathbf{\Sigma}^{-1}\mathbf{B}$ is diagonal. If the diagonal elements of $\mathbf{\Gamma}_0$ are assumed distinct and are ranked $(\gamma_{11} > \dots > \gamma_{NN})$, then $\mathbf{B}$ is unique (admitted).

Estimation by PCA

This method does not require assumptions on the distribution of the observations. It can be applied with the variance matrix or with the correlation matrix.

Let

$$\hat{\lambda}_1 \geq \hat{\lambda}_2 \geq \dots \geq \hat{\lambda}_N$$

some estimators of the eigenvalues of $\mathbf{\Omega}$ and let $\hat{\mathbf{e}}_i$, $i = 1, \dots, N$, the corresponding estimated eigenvectors.

The number of factors being K , the matrix of factors loadings $\mathbf{B}$ is estimated by

$$\hat{\mathbf{B}} = (\hat{B}_{ij}) = \begin{pmatrix} \sqrt{\hat{\lambda}_1} \hat{\mathbf{e}}_1 & \sqrt{\hat{\lambda}_2} \hat{\mathbf{e}}_2 & \dots & \sqrt{\hat{\lambda}_K} \hat{\mathbf{e}}_K \end{pmatrix}.$$

Estimation by PCA

Indeed, we have

$$\hat{\mathbf{B}}\hat{\mathbf{B}}' = \sum_{i=1}^K \hat{\lambda}_i \hat{\mathbf{e}}_i \hat{\mathbf{e}}_i',$$

which approximates the spectral decomposition

$$\mathbf{\Omega}_t = \sum_{i=1}^N \lambda_i \mathbf{e}_i \mathbf{e}_i'.$$

The factors being orthogonal, from

$$\mathbf{\Omega} = \mathbf{B}\mathbf{B}' + \mathbf{\Sigma}$$

an estimator of $\mathbf{\Sigma}$ is

$$\hat{\mathbf{\Sigma}} = \hat{\mathbf{\Omega}} - \hat{\mathbf{B}}\hat{\mathbf{B}}'.$$

Factors rotation

We have seen that the factors are only determined **up to a rotation** (multiplication by an orthogonal matrix) :

$$\mathbf{B}^* = \mathbf{B}\mathbf{C}.$$

It can be interesting to choose this orthogonal matrix $\mathbf{C}$ so as to **discriminate** as much as possible between the **components of $\mathbf{y}$** .

For instance, the "varimax" procedure [Kaiser (1958)], consists in **maximizing** the sum of the **"variances" of the squared loadings**.

The idea is to obtain the maximal dispersion among the entries

$$B_{1j}, B_{2j}, \dots, B_{Nj}$$

of the columns of $\mathbf{B}^*$ (i.e., for each factor j to discriminate as much as possible between the N returns)

Varimax procedure for selecting $\mathbf{B}^* = (B_{ij}^*)$

$c_i^2 = B_{i1}^2 + \dots B_{iK}^2$: part of $V(y_{it})$ explained by the factors.

c_i^2 is **independent of the chosen orthogonal transformation**.

The varimax procedure selects the matrix $\mathbf{B}^*$ such that

$$V = \sum_{j=1}^K \left\{ \frac{1}{N} \sum_{i=1}^N \tilde{B}_{ij}^{*4} - \left(\frac{1}{N} \sum_{i=1}^N \tilde{B}_{ij}^{*2} \right)^2 \right\}$$

is maximal, where $\tilde{B}_{ij}^* = B_{ij}^* / c_i$.

[Normalisation to account for the fact that the c_i can be very different among the variables.]

The procedure often produces **groups of variables** for which, for a given factor, the factor loadings are large or negligible.

Estimation by ML

Under normality assumptions, the parameters of the model can be estimated by Maximum Likelihood.

For instance $\mathbf{B}$ and Σ can be estimated under the assumption of orthogonal factors, which entails $\Omega = \mathbf{B}\mathbf{B}' + \Sigma$, and by assuming that $\mathbf{B}'\Sigma^{-1}\mathbf{B}$ is diagonal.

The likelihood can be maximized using numerical procedures (such as the EM algorithm).

The consistency and asymptotic normality of the ML estimators can be established, which can be used for tests (see Anderson, 1984).

Estimation of the factors (K known)

Suppose that in the factor model $\mathbf{y}_t = \mathbf{c} + \mathbf{B}\mathbf{f}_t + \mathbf{u}_t$,

$$\begin{pmatrix} \mathbf{f}_t \\ \mathbf{u}_t \end{pmatrix} \sim \mathbf{N}\left(\begin{pmatrix} \mathbf{0} \\ \mathbf{0} \end{pmatrix}, \begin{pmatrix} \mathbf{\Gamma} & \mathbf{0} \\ \mathbf{0} & \mathbf{\Sigma} \end{pmatrix}\right).$$

Then

$$\begin{pmatrix} \mathbf{y}_t \\ \mathbf{f}_t \end{pmatrix} \sim \mathbf{N}\left(\begin{pmatrix} \mathbf{c} \\ \mathbf{0} \end{pmatrix}, \begin{pmatrix} \mathbf{B}\mathbf{\Gamma}\mathbf{B}' + \mathbf{\Sigma} & \mathbf{B}\mathbf{\Gamma} \\ \mathbf{\Gamma}\mathbf{B}' & \mathbf{\Gamma} \end{pmatrix}\right).$$

► Gaussian vectors

It follows that the theoretical regression of $\mathbf{f}_t$ on $\mathbf{y}_t$ is

$$E(\mathbf{f}_t | \mathbf{y}_t) = \mathbf{\Gamma}\mathbf{B}'(\mathbf{B}\mathbf{\Gamma}\mathbf{B}' + \mathbf{\Sigma})^{-1}(\mathbf{y}_t - \mathbf{c}).$$

Replacing the unknown parameters by their estimators, we thus obtain an approximation of the factors $\mathbf{f}_t$.

Empirical Application

PCA and Factor Analysis on monthly prices of $N = 5$ stocks,

IBM, Hewlett-Packard (HP), Intel,
Merril Lynch (ML), Morgan Stanley (MS)

observed on the period January 1900 to December 1999 ($n = 120$).

PCA

Empirical covariance and correlation matrices :

$$\hat{\Omega}_n = \begin{pmatrix} 73.10 & & & & \\ 36.48 & 103.60 & & & \\ 27.08 & 48.86 & 113.96 & & \\ 16.06 & 37.59 & 27.96 & 105.56 & \\ 16.33 & 40.72 & 26.86 & 85.47 & 109.91 \end{pmatrix},$$

$$\hat{\rho}_n = \begin{pmatrix} 1 & & & & \\ 0.42 & 1 & & & \\ 0.30 & 0.45 & 1 & & \\ 0.18 & 0.36 & 0.26 & 1 & \\ 0.18 & 0.38 & 0.24 & 0.79 & 1 \end{pmatrix}$$

PCA

Table – Eigenvectors are in columns. Source : Tsay (2002).

	(a) Using the variance matrix				
Eigenvalue	256.16	116.14	64.91	46.82	22.11
Proportion	0.506	0.229	0.128	0.093	0.044
Cumulated	0.506	0.736	0.864	0.956	1.000
Eigenvector	0.246	0.327	0.586	-0.700	0.018
	0.461	0.360	0.428	0.687	-0.050
	0.409	0.585	-0.683	-0.153	0.033
	0.522	-0.452	-0.082	-0.115	-0.710
	0.536	-0.467	-0.036	-0.042	0.701
	(b) Using the correlation matrix				
Eigenvalue	2.456	1.145	0.699	0.495	0.205
Proportion	0.491	0.229	0.140	0.099	0.041
Cumulated	0.491	0.720	0.860	0.959	1.000
Eigenvector	0.342	0.525	0.691	-0.362	-0.012
	0.474	0.314	-0.043	0.820	0.050
	0.387	0.405	-0.717	-0.414	-0.034
	0.503	-0.481	0.052	-0.147	0.701
	0.505	-0.481	0.071	-0.062	-0.711

Main findings

- The 1st component is a linear combination of the 5 stocks with almost equal weights : a **market** component.
- The linear comb. of the components of $\mathbf{y}_t$ with maximal variance (and unit norm) is :

$$0.246\text{IBM}_t + 0.461\text{HP}_t + 0.409\text{INT}_t + 0.522\text{ML}_t + 0.536\text{MS}_t$$

- The first two components represent more than 70% of the total variance. The 2nd comp. opposes 2 sectors : and **industrial** component.
- The 3rd comp. opposes IBM and Intel (more visible on the correlation matrix).

(a) Factor Analysis on the corr. matrix with 2 factors using PCA					
Variable	Factor loadings		Transformed		Explained var.
	f_1	f_2	f_1^*	f_2^*	
IBM	0.536	0.561	0.011	0.776	0.602
HP	0.744	0.335	0.317	0.752	0.665
INTEL	0.607	0.433	0.151	0.730	0.556
ML	0.788	-0.515	0.928	0.158	0.887
MS	0.791	-0.514	0.930	0.161	0.891
Variance	2.456	1.145	1.850	1.751	3.601
Proportion	0.491	0.229	0.370	0.350	0.720
(b) Factor Analysis on the corr. matrix with 2 factors using ML					
Variable	f_1		f_2^*		
	f_1	f_2	f_1^*	f_2^*	
IBM	0.191	0.496	0.087	0.524	0.282
HP	0.394	0.689	0.247	0.755	0.630
INTEL	0.250	0.511	0.141	0.551	0.323
ML	0.800	0.072	0.769	0.232	0.645
MS	0.994	-0.015	0.976	0.186	0.988
Variance	1.881	0.987	1.632	1.236	2.868
Proportion	0.376	0.197	0.326	0.247	0.574

Comments

- $\hat{\mathbf{B}} = (\hat{B}_{ij}) = \left(\sqrt{\hat{\lambda}_1} \hat{\mathbf{e}}_1 \quad \sqrt{\hat{\lambda}_2} \hat{\mathbf{e}}_2 \right) \Rightarrow \hat{\mathbf{B}}\hat{\mathbf{B}}' = \hat{\lambda}_1 \hat{\mathbf{e}}_1 \hat{\mathbf{e}}_1' + \hat{\lambda}_2 \hat{\mathbf{e}}_2 \hat{\mathbf{e}}_2'$
 $\hat{\lambda}_j = \hat{B}_{1j}^2 + \dots + \hat{B}_{5j}^2$: part of the var. explained by the j th factor
 $\hat{B}_{i1}^2 + \hat{B}_{i2}^2$: part of the var. of the i th variable explained by the factors
 $\text{Tr}(\hat{\mathbf{B}}\hat{\mathbf{B}}') = \sum_{i,j} \hat{B}_{ij}^2 = \hat{\lambda}_1 + \hat{\lambda}_2$: explained variance
- More explained variability using PCA
- Using the rotation, the 2 methods find the same groups :
industrial sector vs banking sector
- Rotation does not much change the eigenvalues obtained by ML (very different for PCA).

End of Chapter 4

Proof

Let A and B two **symmetric** $N \times N$ matrices with A **positive definite**.
Then there exists P **invertible** and D **diagonal** such that

$$P'AP = I_N, \quad P'BP = D$$

[since $A^{-1/2}BA^{-1/2}$ is symmetric, there is an orthogonal matrix S such that $S'A^{-1/2}BA^{-1/2}S = D$ with D diagonal. Take $P = A^{-1/2}S$.] Thus

$$A = (P')^{-1}P^{-1}, \quad B = (P')^{-1}DP^{-1}, \quad A^{-1}B = PP'(P')^{-1}DP^{-1} = PDP^{-1},$$

$$\text{Tr}(A^{-1}B) = \sum_{i=1}^N d_i, \quad |A^{-1}B| = \prod_{i=1}^N d_i.$$

If B is semi-pos. def., the d_i 's are nonnegative and

$$\text{Tr}(A^{-1}B) - \log|A^{-1}B| = \sum_{i=1}^N \underbrace{d_i - \log(d_i)}_{\geq 1}$$

It follows that $\text{Tr}(A^{-1}B) - \log|A^{-1}B|$ is **minimal for** $A = B$.

Conditional distributions of the components of a Gaussian vector

Let

$$\begin{pmatrix} X \\ Y \end{pmatrix} \rightsquigarrow \mathcal{N} \left(\begin{pmatrix} \mu_X \\ \mu_Y \end{pmatrix}, \begin{pmatrix} \Sigma_{XX} & \Sigma_{XY} \\ \Sigma_{YX} & \Sigma_{YY} \end{pmatrix} \right).$$

Then the distribution of X conditional on $Y = y$ is

$$\mathcal{N} \left(\mu_X + \Sigma_{XY} \Sigma_{YY}^{-1} (y - \mu_Y), \quad \Sigma_{XX} - \Sigma_{XY} \Sigma_{YY}^{-1} \Sigma_{YX} \right).$$

► Return